

Windows Update Repair

Covers: Update scan failures, "Can't connect to update service," endless "Please wait," stuck downloads, update loop after reboot, post-feature-upgrade hangs.

Decision Tree

```
Windows Update not working
■■■ Shows "Please wait" indefinitely
■   ■■■ Recently upgraded (feature update)? → SCENARIO C
■   ■■■ No recent upgrade → SCENARIO A
■■■ "Can't connect to the update service"
■   ■■■ Can you reach the internet? (ping 8.8.8.8)
■   ■   ■■■ NO → Fix network first
■   ■   ■■■ YES → Can you resolve DNS? (nslookup update.microsoft.com)
■   ■       ■■■ NO → Flush DNS, check DNS server config
■   ■       ■■■ YES → WSUS residue or proxy issue → SCENARIO B
■■■ Update downloads but fails, retries every reboot → SCENARIO D
■■■ Specific error code displayed → Search the code before running generic fixes
```

SCENARIO A: Standard Reset

Orchestrator:

```
.\Invoke-HelpDeskOrchestrator.ps1 -RepairWindowsUpdate
```

Manual (PowerShell):

```
Stop-Service wuauserv, bits, cryptsvc, msiserver -Force
Rename-Item "C:\Windows\SoftwareDistribution" "SoftwareDistribution.old" -Force
Rename-Item "C:\Windows\System32\catroot2" "catroot2.old" -Force
Start-Service cryptsvc, bits, wuauserv, msiserver
(New-Object -ComObject Microsoft.Update.AutoUpdate).DetectNow() # reliable scan trigger; UseClient
```

Manual (CMD):

```
net stop wuauserv
net stop bits
net stop cryptsvc
net stop msiserver
ren "%systemroot%\SoftwareDistribution" SoftwareDistribution.old
ren "%systemroot%\System32\catroot2" catroot2.old
net start cryptsvc
net start bits
net start wuauserv
net start msiserver
:: trigger a scan (equivalent COM call via PowerShell one-liner)
powershell -NoProfile -Command "(New-Object -ComObject Microsoft.Update.AutoUpdate).DetectNow()"
```

Check after 2-3 minutes. Still broken -> Scenario B.

Note (Windows 11 24H2+): if component repair keeps failing, Settings > System >

Recovery > "Fix problems using Windows Update" reinstalls the current build in place while keeping apps, files, and settings. This is Microsoft's supported last-resort repair before an in-place upgrade from ISO.

SCENARIO B: WSUS Residue / Proxy

Machines previously managed by WSUS often retain registry keys pointing to a dead server.

Orchestrator:

```
.\Invoke-HelpDeskOrchestrator.ps1 -RepairWindowsUpdate -ClearLegacyWsusPolicy -ResetWindowsUpdateWin
```

Manual (PowerShell):

```
Remove-ItemProperty "HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate" -Name "WUSever" -Error
Remove-ItemProperty "HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate" -Name "WUStatusServer"
Remove-ItemProperty "HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU" -Name "UseWUSever"
netsh winhttp reset proxy
Restart-Service wuauerv
(New-Object -ComObject Microsoft.Update.AutoUpdate).DetectNow() # reliable scan trigger; UseClient
```

Manual (CMD):

```
reg delete "HKLM\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate" /v WUSever /f
reg delete "HKLM\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate" /v WUStatusServer /f
reg delete "HKLM\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU" /v UseWUSever /f
netsh winhttp reset proxy
net stop wuauerv & net start wuauerv
powershell -NoProfile -Command "(New-Object -ComObject Microsoft.Update.AutoUpdate).DetectNow()"
```

(reg delete returns an error if the value is already absent - that is harmless here.)

SCENARIO C: Post-Feature-Update Hang

1. Run Scenario A first.
2. Check for pending reboot:

```
Get-ItemProperty "HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Component Based Servicing" -Name "
```

CMD:

```
reg query "HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Component Based Servicing\RebootPending"
:: "ERROR: The system was unable to find..." = no reboot pending (good)
```

3. If pending, reboot first, then retry.
 4. Still stuck → Run Windows Update troubleshooter from Settings.
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SCENARIO D: Failed Update Retries Every Reboot

1. Identify the failing update:

```
Get-WindowsUpdateLog
```

2. Hide the problematic update:

```
Install-Module PSWindowsUpdate -Force  
Hide-WindowsUpdate -KBArticleID "KB5XXXXXX" -Confirm:$false
```

3. If it's a required security update, download manually from catalog.update.microsoft.com and install:

```
wusa.exe C:\path\to\update.msu /quiet /norestart
```

Escalation

If Scenarios A-D fail:

- `DISM /Online /Cleanup-Image /RestoreHealth`
- `sfc /scannow`
- In-place upgrade (mount ISO, run setup.exe, select "Keep everything")
- Reimage as last resort

Windows Endpoint Remediation Toolkit — Author: Adrian Melendez

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